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SYSTEM FOR AUTOMATED ASSEMBLY OF A TOOL STRING

Abstract

A system for automated assembly of a tool string may include a pod defining a pod interior, a magazine provided in the pod interior and configured to hold a plurality of perforating guns, a tool string assembler provided in the pod interior and configured to receive the perforating gun and couple the perforating gun to a mid-assembly tool string, a conveyor provided in the pod interior and configured to movably support the mid-assembly tool string, a ramp provided in the pod interior and configured to receive and store a completed tool string from the conveyor, and an ejector sled provided in the pod interior and configured to eject the completed tool string through a tool string ejection port to an exterior of the pod.

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Background/Summary

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/559,649 filed Feb. 29, 2024, the contents of which are incorporated herein by reference.

BACKGROUND

[0002] Assembling a tool string for hydrocarbon extraction can be a dangerous and time-consuming operation. Accidents during tool string assembly can result in serious injury or death. Additionally, completed tool strings may be damaged while being stored until used. Accordingly, it may be desirable to develop a system that can automatically assemble and store tool strings for use at a wellbore site.

[0003] At least an embodiment of a system for automated assembly of a tool string may include a pod defining a pod interior, a magazine provided in the pod interior and configured to hold a plurality of perforating guns, a tool string assembler provided in the pod interior and configured to receive the perforating gun and couple the perforating gun to a mid-assembly tool string, a conveyor provided in the pod interior and configured to movably support the mid-assembly tool string, a ramp provided in the pod interior and configured to receive and store a completed tool string from the conveyor, and an ejector sled provided in the pod interior and configured to eject the completed tool string through a tool string ejection port to an exterior of the pod.

[0004] At least an embodiment of a method for automated assembly of a tool string may include providing a pod defining a pod interior, receiving a tool string component at a tool string assembler disposed in the pod interior, using the tool string assembler to couple the perforating gun to a mid-assembly tool string, conveying the mid-assembly tool string with a conveyor to make room for another component. The steps of receiving a tool string component, using the tool string assembler to couple the tool string component to a mid-assembly tool string, and the conveying the mid-assembly tool string may be iteratively performed until the mid-assembly tool string becomes a completed tool string.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0005] A more particular description will be rendered by reference to exemplary embodiments that are illustrated in the accompanying figures. Understanding that these drawings depict exemplary embodiments and do not limit the scope of this disclosure, the exemplary embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0006] FIG. 1 illustrates perspective views of pods according to an exemplary embodiment.

[0007] FIG. 2 illustrates an interior of a pod according to an exemplary embodiment.

[0008] FIG. 3 illustrates an interior of a pod according to an exemplary embodiment.

[0009] FIG. 4 is a cross-section view of a pod according to an exemplary embodiment.

[0010] FIG. 5 is an end view of a pod according to an exemplary embodiment.

[0011] FIG. 6 is a cross-section view of a pod at line A of FIG. 4 according to an exemplary embodiment.

[0012] FIG. 7 is a cross-section view of a pod at line B of FIG. 4 according to an exemplary embodiment.

[0013] FIG. 8 is a cross-section view of a pod at line C of FIG. 4 according to an exemplary embodiment.

[0014] FIG. 9 is an end view of a pod according to an exemplary embodiment.

[0015] Various features, aspects, and advantages of the exemplary embodiments will become more

apparent from the following detailed description, along with the accompanying drawings in which like numerals represent like components throughout the figures and detailed description. The various described features are not necessarily drawn to scale in the drawings but are drawn to aid in understanding the features of the exemplary embodiments.

[0016] The headings used herein are for organizational purposes only and are not meant to limit the scope of the disclosure or the claims. To facilitate understanding, reference numerals have been used, where possible, to designate like elements common to the figures.

DETAILED DESCRIPTION

[0017] Reference will now be made in detail to various exemplary embodiments. Each example is provided by way of explanation and is not meant as a limitation and does not constitute a definition of all possible embodiments. It is understood that reference to a particular “exemplary embodiment” of, e.g., a structure, assembly, component, configuration, method, etc. includes exemplary embodiments of, e.g., the associated features, subcomponents, method steps, etc. forming a part of the “exemplary embodiment.”

[0018] FIGS. **1-9** show an exemplary embodiment of a system for automated assembly of a tool string. The system may include a pod that contains other components of the system. FIG. **1** shows examples of the pod. In FIG. **1**, the pod may be embodied as a container **102**, such as a shipping container. Alternatively, the pod may be embodied as a trailer **104** that can be towed by a vehicle. In another exemplary embodiment, the pod may be incorporated into a wheeled vehicle such as a truck. In an exemplary embodiment, whether the pod is the container **102** or the trailer **104**, it will be understood that the pod is mobile and can be transported directly to the wellsite. The pod may also include a door such as a door **106** shown on the trailer **104**. The door **106** may allow access to the pod by a worker. As a safety measure, the door **106** may be connected to the rest of the machinery inside the pod so as to automatically shut down all processing and all machines if the door **106** is opened during operation. In an exemplary embodiment, the pod may be climate controlled and sealed from the exterior environment, so as to protect the tool strings and components from contamination and/or environmental damage. In an exemplary embodiment, the pod may be a clean room environment.

[0019] FIG. **2** shows an exemplary embodiment of a pod interior **202**. As seen in FIG. **2**, the system may include a magazine **204**, an end cap remover **210**, a tool string assembler **212**, a conveyor **216**, and a ramp **218** disposed within the pod interior **202**. The magazine **204** may be configured to hold a plurality of perforating guns **206**. In an exemplary embodiment, the magazine **204** may be configured as a vertically oriented conveyor system. The magazine **204** may rotate, revolve, or otherwise sequentially convey each of the plurality of perforating guns **206** to a magazine ejection point **208**. The perforating guns **206** may be supplied to the magazine **204** from an exterior of the pod, as discussed further herein.

[0020] At the magazine ejection point **208**, one of the perforating guns **206** is transferred from the magazine **204** to the end cap remover **210**. In an exemplary embodiment, the perforating guns **206** may be shipped with end caps fastened at one or both ends to protect internal components and keep out debris. The end cap remover **210** may be configured to secure the perforating gun **206** and remove the end caps from the perforating gun **206**. After removal, the end caps may be deposited in an end cap collector bin **510** (see FIG. **6**) for storage.

[0021] After removal of the end caps, the perforating gun **206** may be transferred to the tool string assembler **212**. The tool string assembler **212** is configured to sequentially connect tool string components such as a top connector, the perforating guns **206**, a setting tool, or a release tool. It will be understood that the system is not limited to these tools, and that other tool string tools such as weight bars or any other suitable tool may be used in the automatic assembly system. If the tool string component is the first component on the tool string, then it may be delivered to the tool string assembler **212** and placed on a conveyor **216**, where it becomes the first component of a mid-assembly tool string **214**. An assembly sled **302** (see FIG. **3**) may be coupled to the mid-assembly

tool string **214**. The assembly sled **302** may be connected to an assembly rail **304** (see FIG. 3) and may be configured to pull the mid-assembly tool string **214** along the conveyor **216** to make room for the next component in the tool string assembler **212**.

[0022] Once the mid-assembly tool string **214** is established on the conveyor, the next component, such as a perforating gun **206**, may be delivered to the tool string assembler **212**. The tool string assembler **212** may clamp an end of the mid-assembly tool string **214** so that the next component can be attached. The next component may be attached by rotating the component relative to the mid-assembly tool string **214** so as to engage threads and screw the component to the mid-assembly tool string **214**. In an alternative embodiment, the component may be engaged with the mid-assembly tool string **214** by pushing, bayonet-style attachment, twisting, or other suitable method of connection.

[0023] In an exemplary embodiment, the assembly sled **302** may include an electrical tester. The electrical tester may be in electrical communication with a top connector of the mid-assembly tool string **214**. The electrical tester may be configured to test electrical connections between each of the components in the mid-assembly tool string **214**. For example, the electrical tester may be configured to test line-in, line-out, and ground of each component as it is attached to the mid-assembly tool string **214**. If any components fail the electrical testing, the mid-assembly tool string **214** can be removed from production so that a faulty tool string is not deployed to the wellbore.

[0024] In an exemplary embodiment, the mid-assembly tool string **214** may be assembled in a top-down fashion. In other words, the first component of the mid-assembly tool string **214** may be a top connector, and then subsequent components are added until the bottom of the tool string is reached. The method facilitates the use of an electrical tester on the assembly sled **302**, as the electrical tester can easily connect to the top connector and run the necessary tests. However, it will also be understood that in some embodiments, it may be desirable to assemble the tool string from the bottom-up, i.e., starting with the bottommost component and ending with the top connector.

[0025] A user **404** may program the system to assemble a predetermined number and type of tool string components in order to form a completed tool string. For example, a completed tool string may include one or more of a top connector, perforating gun(s), setting tool, and or release tool. It will be understood that the tool string components are not limited to this list and other tool string components may be used as appropriate. Once the tool string assembler **212** attaches the predetermined number and type of tool string components to the mid-assembly tool string **214**, the mid-assembly tool string **214** becomes a completed tool string **220**. To achieve the completed tool string **220**, the system may iteratively perform steps of the receiving a tool string component, using the tool string assembler **212** to couple the tool string component to a mid-assembly tool string, and the conveying the mid-assembly tool string until the completed tool string **220** is achieved.

[0026] Once the mid-assembly tool string **214** is completed, it may be moved to a ramp **218**. The ramp **218** may be configured to store one or more completed tool strings **220** until operators are ready to use one of the completed tool strings **220**. The ramp **218** may be configured so that the completed tool strings **220** are gravity fed to a point where they can be transferred to an exterior of the pod by an ejector sled **306**. The ejector sled **306** may be coupled to an ejector rail **308** and configured to push a bottommost completed tool string **220** out of the pod through a tool string ejection port **402** (see FIG. 4). It will be understood that the completed tool strings **220** may be stored within the protected environment of the pod up until the moment they are needed. In this way, the completed tool strings **220** may be protected from environmental damage or accidental damage on the work site.

[0027] As seen in FIG. 4 and FIG. 5, one or more users **404** may be positioned outside of the trailer **104** to supply perforating guns **206** and/or other tool string components to the system. For example, a plurality of perforating guns **206** may be stored in a component storage **406** stored outside of the trailer **104**. The user **404** may manually transfer the perforating guns **206** from the component storage **406** and insert them through a perforating gun insertion port **502** to load the magazine **204**

inside the trailer **104**. Similarly, components such as a top connector, a setting tool, or a release tool may also be stored within a component storage **406**. The user may supply a top connector through a top connector insertion port **504**, a setting tool through a setting tool insertion port **506**, or a release tool through a release tool insertion port **508**, where they can be processed by the tool string assembler **212** and added to the mid-assembly tool string **214**.

[0028] In an alternative embodiment, instead of a user, automatic transfer systems may be used to selected components from one or more component storages **406** and supply them to the trailer **104** through the respective insertion ports. In this scenario, the various components (or alternatively, the component storage **406**) may be equipped with barcodes and/or RFID tags or the like, and the system may include scanners and/or readers capable of reading such barcodes or RFID tags. This would facilitate the automation of the feeding system and help to insure that the proper components are being selected.

[0029] FIGS. **6-8** show additional views of the trailer **104** at various cross-section points.

[0030] This disclosure, in various embodiments, configurations and aspects, includes components, methods, processes, systems, and/or apparatuses as depicted and described herein, including various embodiments, sub-combinations, and subsets thereof. This disclosure contemplates, in various embodiments, configurations and aspects, the actual or optional use or inclusion of, e.g., components or processes as may be well-known or understood in the art and consistent with this disclosure though not depicted and/or described herein.

[0031] The phrases “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

[0032] In this specification and the claims that follow, reference will be made to a number of terms that have the following meanings. The terms “a” (or “an”) and “the” refer to one or more of that entity, thereby including plural referents unless the context clearly dictates otherwise. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein.

Furthermore, references to “one embodiment”, “some embodiments”, “an embodiment” and the like are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related.

Accordingly, a value modified by a term such as “about” is not to be limited to the precise value specified. In some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Terms such as “first,” “second,” “upper,” “lower” etc. are used to identify one element from another, and unless otherwise specified are not meant to refer to a particular order or number of elements.

[0033] As used herein, the terms “may” and “may be” indicate a possibility of an occurrence within a set of circumstances; a possession of a specified property, characteristic or function; and/or qualify another verb by expressing one or more of an ability, capability, or possibility associated with the qualified verb. Accordingly, usage of “may” and “may be” indicates that a modified term is apparently appropriate, capable, or suitable for an indicated capacity, function, or usage, while considering that in some circumstances the modified term may sometimes not be appropriate, capable, or suitable. For example, in some circumstances an event or capacity can be expected, while in other circumstances the event or capacity cannot occur—this distinction is captured by the terms “may” and “may be.”

[0034] As used in the claims, the word “comprises” and its grammatical variants logically also subtend and include phrases of varying and differing extent such as for example, but not limited thereto, “consisting essentially of” and “consisting of.” Where necessary, ranges have been

supplied, and those ranges are inclusive of all sub-ranges therebetween. It is to be expected that the appended claims should cover variations in the ranges except where this disclosure makes clear the use of a particular range in certain embodiments.

[0035] The terms “determine”, “calculate” and “compute,” and variations thereof, as used herein, are used interchangeably and include any type of methodology, process, mathematical operation or technique.

[0036] This disclosure is presented for purposes of illustration and description. This disclosure is not limited to the form or forms disclosed herein. In the Detailed Description of this disclosure, for example, various features of some exemplary embodiments are grouped together to representatively describe those and other contemplated embodiments, configurations, and aspects, to the extent that including in this disclosure a description of every potential embodiment, variant, and combination of features is not feasible. Thus, the features of the disclosed embodiments, configurations, and aspects may be combined in alternate embodiments, configurations, and aspects not expressly discussed above. For example, the features recited in the following claims lie in less than all features of a single disclosed embodiment, configuration, or aspect. Thus, the following claims are hereby incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment of this disclosure.

[0037] Advances in science and technology may provide variations that are not necessarily express in the terminology of this disclosure although the claims would not necessarily exclude these variations.

Claims

1. A system for automated assembly of a tool string, the system comprising: a pod defining a pod interior; a magazine provided in the pod interior and configured to hold a plurality of perforating guns; a tool string assembler provided in the pod interior and configured to receive the perforating gun and couple the perforating gun to a mid-assembly tool string; a conveyor provided in the pod interior and configured to movably support the mid-assembly tool string; a ramp provided in the pod interior and configured to receive and store a completed tool string from the conveyor; and an ejector sled provided in the pod interior and configured to eject the completed tool string through a tool string ejection port to an exterior of the pod.
2. The system of claim 1, further comprising: an end cap remover provided in the pod interior and configured to receive a perforating gun from the magazine and remove an end cap from the perforating gun.
3. The system of claim 1, further comprising: an assembly sled coupled to the mid-assembly tool string and configured to move the mid-assembly tool string along the conveyor.
4. The system of claim 1, wherein: the mid-assembly tool string comprises a top connector; and the assembly sled comprises an electrical tester configured to be in electrical communication with the top connector to test electrical connections within the mid-assembly tool string.
5. The system of claim 1, further comprising a perforating gun insertion port configured to receive perforating guns from the exterior of the pod and store the perforating guns in the magazine.
6. The system of claim 1, further comprising a top connector insertion port configured to receive a top connector from the exterior of the pod and deliver the top connector to the tool string assembler.
7. The system of claim 1, further comprising a setting tool insertion port configured to receive a setting tool from the exterior of the pod and deliver the setting tool to the tool string assembler.
8. The system of claim 1, further comprising a release tool insertion port configured to receive a release tool from the exterior of the pod and deliver the release tool to the tool string assembler.
9. The system of claim 1, wherein: the pod further comprises a door to the exterior of the pod; and the system is configured to shut down operation if the door is opened.

- 10.** The system of claim 1, wherein the pod is climate controlled and sealed from the exterior of the pod.
- 11.** The system of claim 1, wherein the pod is mobile.
- 12.** The system of claim 11, wherein the pod is a wheeled trailer.
- 13.** A method for automated assembly of a tool string, the method comprising: providing a pod defining a pod interior; receiving a tool string component at a tool string assembler disposed in the pod interior; using the tool string assembler to couple the perforating gun to a mid-assembly tool string; conveying the mid-assembly tool string with a conveyer to make room for another component; wherein the receiving a tool string component, using the tool string assembler to couple the tool string component to a mid-assembly tool string, and the conveying the mid-assembly tool string are iteratively performed until the mid-assembly tool string becomes a completed tool string.
- 14.** The method for automated assembly of claim 13, further comprising transferring the completed tool string from the conveyer to a ramp.
- 15.** The method for automated assembly of claim 13, wherein: the pod comprises a magazine holding a plurality of perforating guns; the tool string component is a perforating gun; and the receiving a tool string component at the tool string assembler comprises receiving a perforating gun from the magazine.
- 16.** The method for automated assembly of claim 13, wherein: an end cap remover is provided in the pod interior; and the method further comprises using the end cap remover to remove an end cap from the perforating gun before receiving the tool string component at the tool string assembler.
- 17.** The method for automated assembly of claim 13, further comprising: testing electrical connections within the mid-assembly tool string.
- 18.** The method for automated assembly of claim 13, wherein: the receiving a tool string component at the tool string assembler comprises receiving a top connector from the exterior of the pod and delivering the top connector to the tool string assembler.
- 19.** The method for automated assembly of claim 13, wherein: the receiving a tool string component at the tool string assembler comprises receiving a setting tool from the exterior of the pod and deliver the setting tool to the tool string assembler.
- 20.** The method for automated assembly of claim 13, further comprising transporting the pod to a wellbore location.
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